

	$T = 310\text{ K}$
16.	Ans: A Process (i): expansion under same pressure will cause the temperature of the gas to increase. Hence, from $\Delta U = Q + W$, internal energy increases, W is negative, hence Q must be positive. Process (ii): expansion under same temperature means that internal energy remains constant. From $0 = Q + W$, since W is negative, hence Q must be positive.
17.	Ans: A